



Study on the acrylate coating with chemically bonded functional group for antiseawater bioadhesion

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Received: 25 March 2021 / Revised: 4 November 2021 / Accepted: 14 November 2021
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Abstract The adhesion foundation of seawater organisms can be eliminated by killing the bacteria on the surface of objects in seawater. Thus, 4-bromo-2-(4-chlorophenyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile (Econea) was selected as the core functional group to conduct the solution grafting reaction with the epoxy-containing acrylate copolymer (EA) to yield the marine antifouling paint, EA–Econea. UV absorption spectrum, Fourier transform infrared spectrum (FTIR) and energy dispersive spectrometer (EDS) analyses verified that Econea was exactly bonded to the EA molecular chain. Furthermore, the antibacterial test, confocal laser scanning microscope (CLSM), and antifouling experiments confirmed that when the mass fraction of Econea in the reaction reaches 0.5%, the coating affords 99.9% antibacterial effect on *Escher-*

ichia coli and *Staphylococcus aureus*. The Econea and EA molecular chain combination cannot be easily broken, which lays the foundation for the long-term antifouling performance of the coating. A stainless steel plate coated with EA–Econea was immersed in a river rich in microorganisms for two months, and the plate was not contaminated.

Keywords Econea, Acrylate copolymer, Marine antifouling paint, Antibacterial

Introduction

Marine organisms, such as barnacles, mussels and algae, can easily attach to the seawater submerged surfaces of marine equipment, such as ships, offshore platforms, offshore wells and docks.^{1,2} Such harmful attachments severely reduce the speed of a ship's navigation, increase fuel consumption, and aggravate the corrosion of hull steel plates and underwater facilities, greatly shortening the service life of marine equipment.³ Additionally, organisms attached to ship surfaces will be carried around the world with the ship, causing the infiltration of new species and destruction of the local ecological balance.⁴ Therefore, the global marine organism pollution will annually cause hundreds of billions of dollars in economic losses.⁵

However, among the current methods to prevent the adhesion of marine organisms, the simplest, most economical, and most effective method is the use of antifouling paint.⁶ However, numerous coatings of this type suffer from toxicity and short life problems.⁷ To change this situation, two kinds of antibioadhesion coatings have received significant attention worldwide: self-polishing coatings and low-surface-tension silicon coatings.^{8,9} However, self-polishing coatings, as the name suggests, continuously fall off, which reduces their service life.¹⁰ In contrast, for low-surface-tension

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silicone coatings, although the low surface tension makes it difficult for marine organisms to adhere to it, the coating easily sheds from the substrate surface as its adhesion to the shell of the ship is difficult.^{11–13} Thus, although there are currently two development directions for coatings, a coating that can simultaneously meet the requirements of antiseawater bioadhesion, nontoxicity, and environmental protection still needs to be developed. A large number of previous studies have shown that the adhesion of bacteria is the first step of seawater bioadhesion.^{14–16} Therefore, this study is based on the assumption that seawater biological adhesion can be blocked by killing bacteria. Simultaneously, to make the coating safe and durable, 4-bromo-2-(4-chlorophenyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile (Econea)¹⁷ is bonded to the matrix resin molecular chain through a chemical reaction.

Econea is an environmentally friendly marine antifouling agent.¹⁸ It was developed to protect plants from insects and mites during their growth. Since it is environmentally friendly, it has since been used as an antifouling paint, and it currently plays an important role in the global marine antifouling system.^{19–22} Silva et al.²³ used isocyanate-modified Econea to yield a marine antifouling coating based on dimethylsiloxane (PDMS), which can keep ships clean for more than nine months of global navigation. However, the adhesion of PDMS coatings to ship hulls is poor, which is a problem that has always hindered the development of PDMS coatings. Moreover, they did not present a clear antifouling mechanism.

Therefore, herein, polyacrylate is selected as the matrix resin of the coatings, and Econea is connected to the molecular chain of polyacrylate through chemical reactions to synthesize a safe, environment friendly, long-lasting, practical and low-cost seawater antibiological adhesion coating and address the global needs.

Experiment section

Material

Methyl methacrylate (MMA, analytical grade), n-butyl acrylate (BA, analytical grade), glycidyl methacrylate (GMA, analytical grade), and azodiisobutyronitrile (AIBN, analytical grade) were purchased from Shanghai Taitan Co. Ltd. Econea (analytical grade) was purchased from Hubei Qifei Pharmaceutical Chemical Co. Ltd. Ethyl acetate (EAC, chemical grade) was purchased from Shanghai Macklin Biochemical Co. Ltd.

Preparation of polyacrylate antibacterial coating

First, MMA, BA, GMA, and EAC were weighed, separately added to a three-mouth flask and stirred into a mixed monomer solution. Then, azodiisobutyronitrile (AIBN) was dissolved into this solution and reacted at 70°C for 6 h to yield polyacrylate with epoxies (EA). Thereafter, the grafting reaction of Econea in the EAC solution onto the EA molecule in the above solution was continued for 4 h, which yielded an acrylate antibacterial paint (EA–Econea). Scheme 1 displays the preparation process of EA–Econea.

Preparation of the antifouling coating board

EA–Econea antifouling paint was brushed on a precoated steel plate, and it was placed in a fume hood to naturally form a film. Then, the steel plate was placed in a mixed solution of deionized water and ethanol and soaked three times (12 h each time) to remove the unreacted Econea and unreactive monomer. Next, the plate was dried at 55°C for 12 h in an oven, affording an antifouling coating board. The preparation complies with ISO 1513-2010.

Analysis and characterization

Appearance analysis

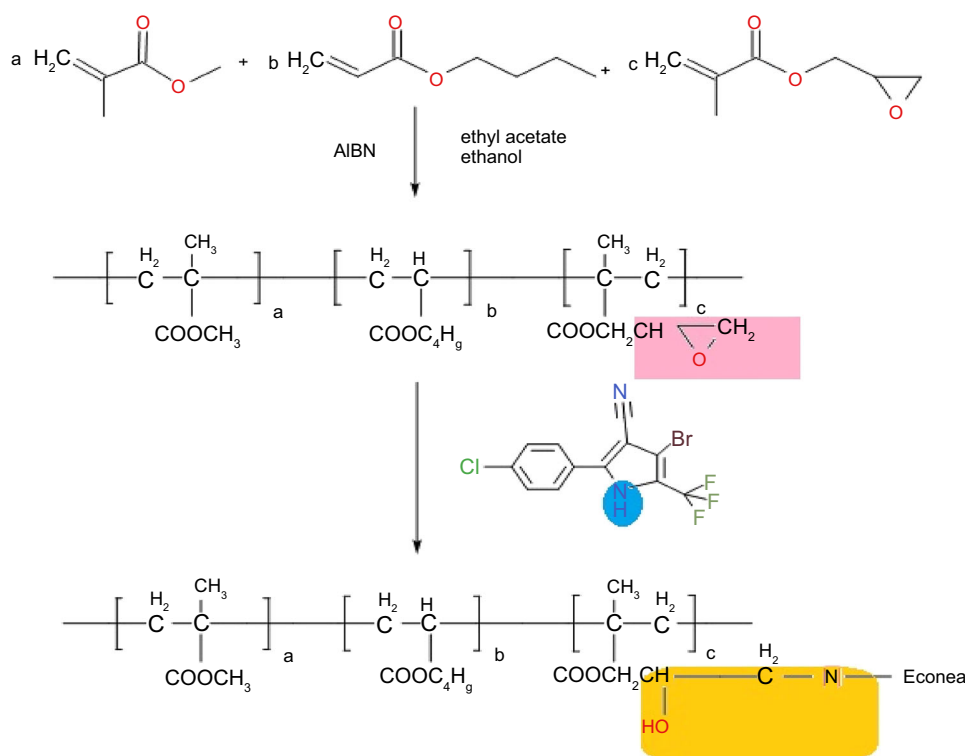
Using the same method as in section [Preparation of polyacrylate antibacterial coating](#), pure polyacrylate resin was obtained by the copolymerization of only MMA and BA, abbreviated as MA. Using EAC as the co-solvent, solutions of EA–Econea and MA/Econea mixtures with the same Econea content were prepared; they were evenly coated on the bottom of a glass sample dish and placed in a fume hood to form two films. The appearance characteristics of the two films were visually observed.

Infrared spectrum analysis

As in section [Preparation of the antifouling coating board](#), a coating was prepared on the bottom of the glass sample dish, and the free monomer and Econea were removed. After being fully dried, the coated film was removed and placed on the sample stage of the Nicolet 6700 FTIR instrument. Inspection was performed in the 4000–400 cm^{−1} range, and scanning was performed 32 times.

UV absorption spectrum analysis

Econea alcoholic solutions with varying Econea concentrations (5, 10, 15, 20 and 25 mg/L) were prepared, and their absorbances at the absorption wavelength of



Scheme 1: Preparation process of EA-Econea

283 nm were detected using an ultraviolet (UV)–visible spectrophotometer (UV-6100). The UV absorption standard curve of Econea in ethanol was obtained. The EA-Econea coating sample painted on the glass surface was removed and ground. Next, the weighed EA-Econea coating sample was extracted using ethanol. The Econea concentration extracted from the EA-Econea coating by alcohol was calculated according to the standard curve and the grafting rate of Econea on polyacrylate was obtained.

EDS analysis

After the film samples were purified and dried, a Falion 60S energy spectrometer was used to qualitatively analyze C, O, N, Cl, Br, and F elements in a randomly selected area with a size of $5 \times 4 \mu\text{m}^2$ in the film samples and their distributions.

Antimicrobial performance test

The selected bacterial species were *Escherichia coli* and *Staphylococcus aureus*. Before the testing, phosphate-buffered saline (PBS) was prepared. The sterilized sample film was cut into a size of $5 \times 5 \text{ cm}^2$ and horizontally placed on a petri dish. Next, 0.1 mL of the bacterial suspension was aspirated with a concentration of 10^6 CFU/mL and added dropwise to the central area of the sample. The sterilized polyethylene film was

then placed on the sample surface to evenly distribute the bacterial liquid on the sample. The petri dish was covered and wrapped with plastic to keep it moist. The petri dishes were placed in an incubator for 24 h; the temperature and humidity were maintained at 37°C and above 90%, respectively. Then, by repeatedly washing the polyethylene film and the sample with 10 mL PBS solution using a pipette, the washing solution was diluted 10 and 100 times, respectively, to obtain a 100-fold diluted bacterial suspension. After that, 0.1 mL of the diluted suspension was pipetted and spread evenly on an agar medium, which was then placed in an incubator at 37°C . After 24 h, the number of viable colonies on the medium was counted. In the same test process, the EA resin film was used as a control sample, and the measurement was performed three times in parallel. The inhibition factor is calculated as follows:

$$\text{Inhibition Factor} = (W_b - W_a) / W_b \times 100\% \quad (1)$$

Here, W_a and W_b denote the number of colonies in the antibacterial and control samples, respectively.

CLSM analysis

First, a fluorescein isothiocyanate (FITC) probe was prepared in the ethyl acetate (EAC) solution with a concentration of 1 mg/mL. *S. aureus* was incubated in the LB broth to the exponential phase, centrifuged to

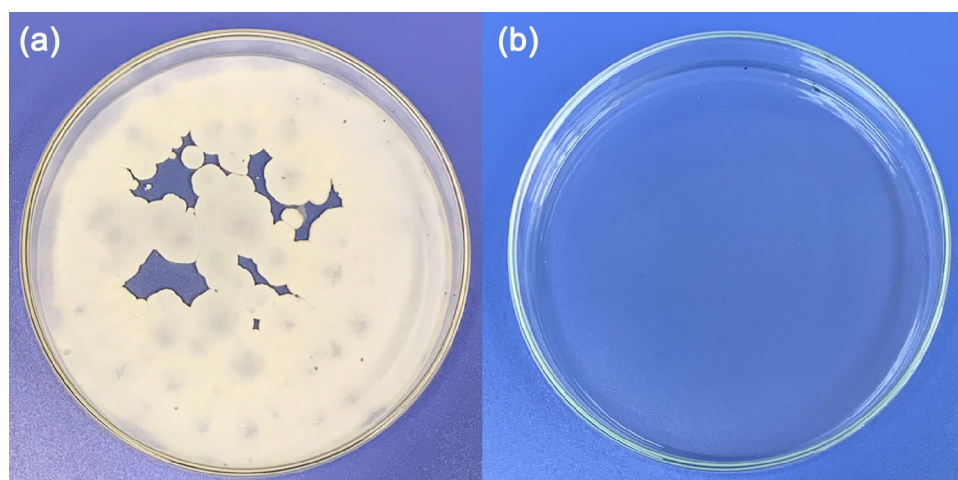


Fig. 1: Pictures of MA/Ecnea mixture (a) and EA-Ecnea (b)

discard the supernatant, washed twice with PBS and resuspended in PBS buffer. Next, 1 mL of FITC solution was added to the bacterial suspension and stained for 120 min in the dark, and then, the FITC solution was filtered out. PBS was added into the bacterial solution to afford a 2.1×10^8 cfu/mL *S. aureus* suspension. Then, the status and fluorescence signal of the bacteria were observed through a laser confocal microscope (Nikon-A1R) after the bacterial suspension was placed in contact with the purified and dried EA-Ecnea coating.

Intact and living bacteria will not fluoresce after staining. Only when a bacteria cell membrane is ruptured, the nonspecific enzyme in the cell will emerge and come in contact with FITC, causing FITC to glow. Whether the cell membrane of the bacteria is broken during the death process can be judged via this test method.

Hydrophilic analysis

As in section [Preparation of the antifouling coating board](#), a coating was prepared on the bottom of the sample dish to yield a flat film, and the free monomer and Ecnea were removed. After fully drying, the coated film was removed and placed on the testing platform of the JC2000D3 contact angle tester (Shanghai Zhongchen Digital Technology Equipment Co., Ltd.) to test the static contact angle of water with the sample. Each sample was tested five times in different areas and the average value was taken.

Antifouling performance test (the film analysis method)

To facilitate the simulation of the antifouling effect in the ocean, river water that was rich in microorganisms, such as bacteria, fungi, and algae, was used to simulate the marine environment. As in section [2.3](#), the steel

plate was coated, and the free monomer and Ecnea were removed. After fully drying, the steel plate was immersed in the river water, and the antifouling effect was continuously observed for 1 day, 7 days, and 60 days.

Results and discussion

Appearance analysis

Figure [1a](#) displays the thin film formed by blending 9.1 wt% of Ecnea and 90.9 wt% of MA using a co-solvent to form a uniform transparent solution and then coating and evaporating the solvent. Obviously, a large amount of uneven white material is present in the coating. This is due to the phase separation of Ecnea from MA after the volatilization of the solvent and indicates that MA and Ecnea are incompatible. Figure [1b](#) displays a photo of the resin film formed due to the chemical reaction of Ecnea and EA with the same mass ratio of Ecnea as in Fig. [1a](#), and the film is also formed by natural evaporation of the solvent. The figure shows that the coating is completely transparent. Additionally, it fully shows that EA and Ecnea form a homogeneous resin after chemical bonding.

FTIR analysis

Figure [2](#) shows the FTIR spectra of Ecnea, EA and purified EA-Ecnea. As shown in the figure, the 2235 cm^{-1} absorption peak is the stretching vibration peak of the aromatic cyanide $\text{C}\equiv\text{N}$, and the absorption peaks at $1350\text{--}1230$ and $1270\text{--}1100\text{ cm}^{-1}$ are the aromatic amine C--N and the aromatic C--F stretching vibration peaks, respectively. The absorption peak at 826 cm^{-1} is the stretching vibration peak of C--Cl . However, these absorption peaks are not present in the

EA spectrum. Contrarily, the stretching vibration peak of the ester carbonyl group can be clearly seen at 1721 cm^{-1} in EA spectrum, while it is not present in the Econeal spectrum.

The comparison of the EA–Econeal and EA FTIR spectra shows that the characteristic absorption peaks of the ester carbonyl group in EA are present in the EA–Econeal spectrum. Additionally, absorption peaks at 2235 and 826 cm^{-1} are present, which do not exist in the FTIR spectrum of EA but exist in that of Econeal. This fully confirms the effective binding of Econeal to the molecular chain of EA.

UV absorption spectrum analysis

A series with varying concentrations of Econeal ethanol solution was prepared, and their absorption spectrum

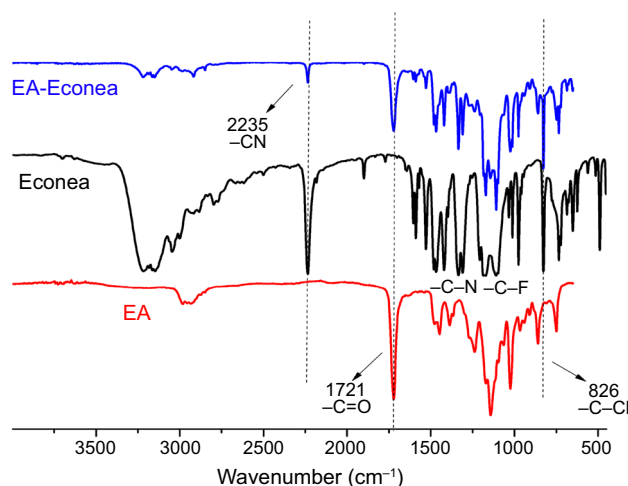
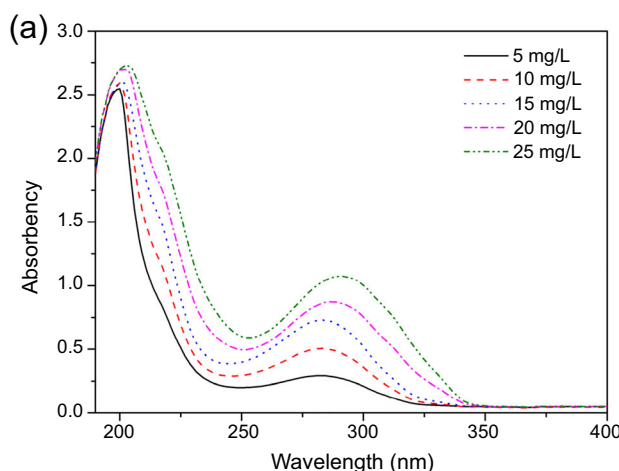


Fig. 2: Infrared spectra of Econeal, EA and EA–Econeal



curves in the $190\text{--}400\text{ nm}$ range were obtained (Fig. 3a). Each spectral curve exhibited an obvious absorption peak near 283 nm , and red shift was observed with increasing concentration. Based on the relationship between the absorbance of the absorption peak ($\text{Abs}_{283\text{nm}}$) and concentration (C_{Econeal}), the scatter plot was composed and linear fitting was performed. The absorbance of Econeal at 283 nm (Fig. 3b) was obtained as follows:

$$\text{Abs}_{283\text{nm}} = 0.03652C_{\text{Econeal}} + 0.13614 \quad (2)$$

The dried EA–Econeal coating on the glass surface was prepared as in [UV absorption spectrum analysis section](#). Then, it was extracted with ethanol for 24 h using a Soxhlet extractor after being peeled from the glass surface, fully pulverized and accurately weighed. Thereafter, the absorbance of the absorption peak near 283 nm was recorded using a UV spectrometer. The free Econeal content was calculated according to Eq. (2). Based on the quantity of the EA–Econeal film before extraction and the ratio of raw materials, the grafting efficiency of Econeal in EA–Econeal was calculated to be 62.2% . This may be because Econeal's reactive groups are secondary amines, which are not as reactive as primary amines; thus, some secondary amine groups do not participate in the reaction.

EDS analysis

EDS was used to analyze the element composition and distribution of the purified EA–Econeal film with a mass fraction of Econeal of 1% (Fig. 4).

As shown in Fig. 4, in addition to large amounts of C and O, N, Cl, F, and Br are evenly distributed on the

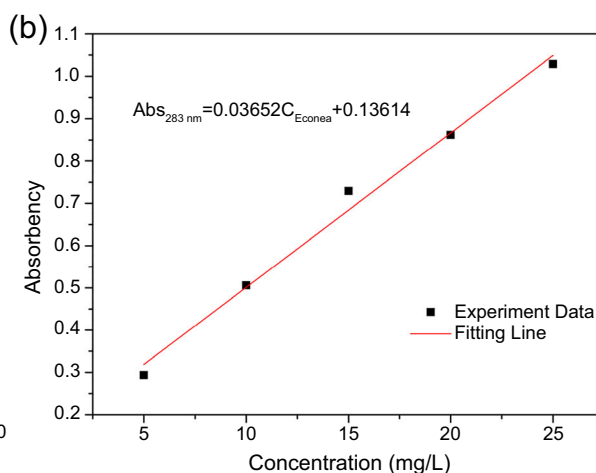


Fig. 3: UV absorption spectra of Econeal ethanol solution (a) and the fitting line of the relationship between UV absorbance and concentration of Econeal (b)

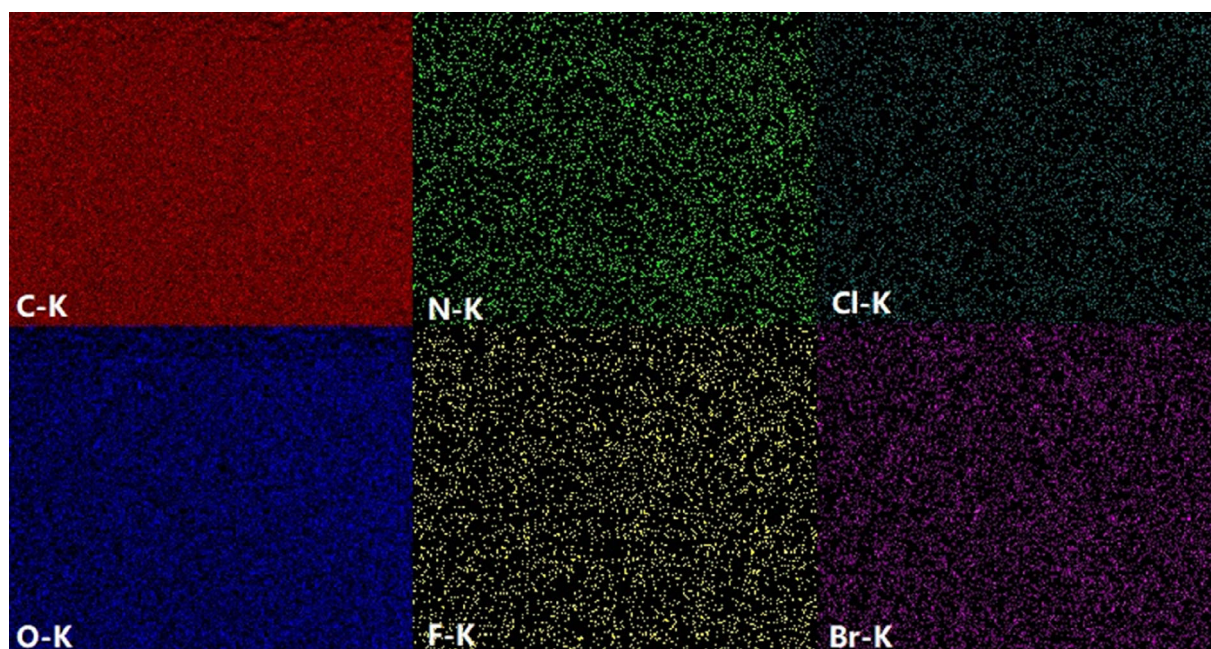


Fig. 4: EDS images of EA-Econeal coating film

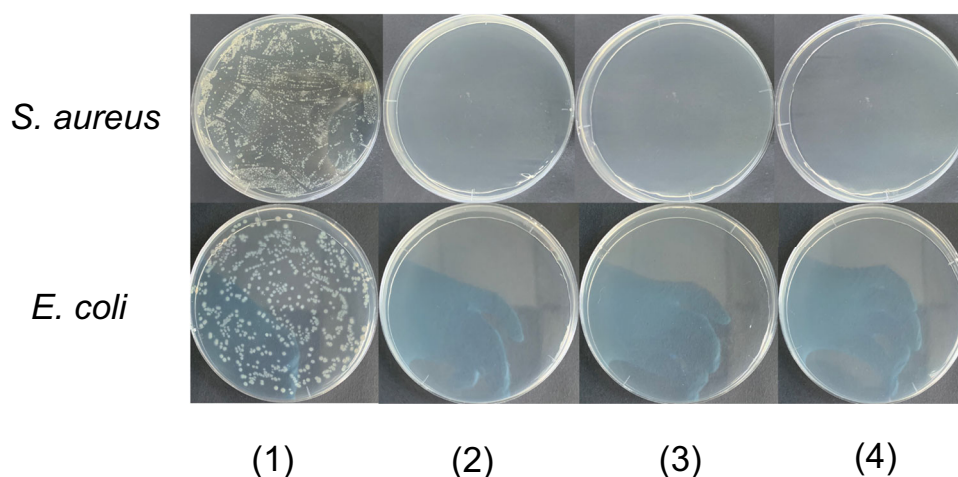


Fig. 5: Antibacterial results using the film analysis method of pure EA film (1) and EA-Econeal film extracted using deionized water (2), ethanol (3), and detergent (4)

EA-Econeal sample surface. Moreover, the Econeal bonded to the EA molecular chain is extremely uniformly distributed in the EA molecular chain and becomes one phase, which well matches the results in Fig. 1.

Antibacterial performance

The inhibition factors of pure EA and EA-Econeal films extracted using different solvents were detected by the film analysis method. The results are shown in Fig. 5. Figure 5(1) displays the antibacterial status of pure EA. The agar plate was full of bacteria, indicating

that EA itself has no antibacterial effect. However, for the EA-Econeal film with a mass fraction of 0.5% Econeal, regardless of the solvent used for extraction, the inhibition rates against *Escherichia coli* and *Staphylococcus aureus* were >99.9%. This implies that the EA-Econeal film possesses not only excellent antibacterial effect and but also extraction resistance. Undoubtedly, the durability is due to the chemical bond of Econeal to the EA molecular chain.

Figure 6 shows the antibacterial effect of EA-Econeal films with different Econeal mass contents. As shown in the figure, the inhibition factor of the material increases with the Econeal content. When the mass content of Econeal reached 0.5 wt%, the

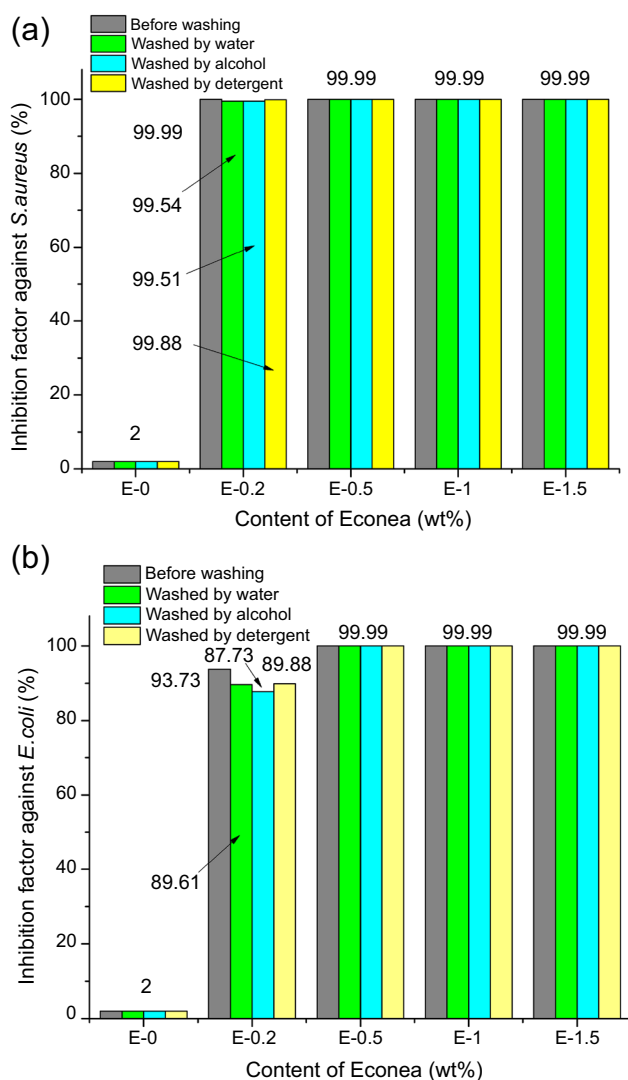


Fig. 6: Bacterial inhibition factors of *E. coli* (a) and *S. aureus* (b) of the EA-Econeal coating films with different Econeal contents before and after extraction with water, ethanol, and detergent

inhibition factor of EA-Econeal film against *E. coli* and *S. aureus* reached over 99.9%. Furthermore, EA-Econeal exhibited stronger antibacterial ability against *S. aureus* than *E. coli*, which is consistent with the literature.¹⁶

Figure 6 also shows that the inhibition factor of the EA-Econeal films before and after extraction with water, ethanol and detergent is almost unchanged after the Econeal mass fraction reaches 0.5%. Both before and after the extraction, the inhibition factor of the coating was >99.9%. This confirms that the EA-Econeal coating film exhibits excellent antibacterial durability, which is obviously inseparable from the bonding of Econeal and EA. However, since Econeal is bonded in the molecular chain of EA and it cannot be released, how exactly does it kill bacteria?

Laser confocal microscopic analysis

Row (a) displays phase contrast microscope images; row (b) displays the fluorescence microscope images; series (1) displays the image of the *S. aureus* suspension after FITC staining; series (2) displays the image of the EA-Econeal film covered with the PBS buffer; and series (3) and (4) display the images of EA-Econeal films in contact with the FITC stained *S. aureus* suspension for 3 and 10 min, respectively.

The FITC probe was used as a fluorescent dye to stain the *S. aureus* suspension with a concentration of 2.1×10^8 cfu/mL, and then, CLSM was used to observe the photos before and after the EA-Econeal film was in contact with the suspension. The results are shown in Fig. 7.

Figure 7(1) displays the image of the *S. aureus* suspension after FITC staining. Numerous bacteria are present in image 7 (a, 1) of the phase contrast microscope, but no fluorescence can be seen in image 7 (b, 1) of the fluorescence microscope. This is because living bacteria have intact cell membranes, and the FITC dye does not contact the nonspecific enzymes in the cell and, therefore, does not emit light.

Figure 7(2) displays the image of the EA-Econeal film surface covered with 10 μ L of the PBS buffer droplets. Of course, no bacteria can be seen in image 7 (a, 2) of the phase contrast microscope, but the fluorescence is also not seen in image in 7 (b, 2), indicating that the EA-Econeal film itself does not fluoresce.

Figure 7(3) displays the image of the EA-Econeal film exposed to FITC stained *S. aureus* suspension for 3 min. Many bacteria are present in image 7 (a, 3), but they appear as imaginary and broken compared to those in image 7 (a, 1). In the fluorescence microscope image 7 (b, 3), fluorescence is clearly displayed, indicating the rupturing of the bacterial cell membrane. It also explains why the bacteria in image 7 (a, 3) appear imaginary and broken.

Figure 7(4) displays an image of the EA-Econeal film exposed to FITC stained *S. aureus* suspension for 10 min. The bacteria in image 7 (a, 4) are more imaginary and broken than those in image 7 (b, 4), but the fluorescence intensity in image 7 (b, 4) is stronger than that in image 7 (a, 4). This fully shows that as the contact time increases, the membranes of more bacteria break.

Thus, Fig. 7 shows that the EA-Econeal film achieves sterilization by destroying the cell membrane of the bacteria, as long as it is in contact with bacteria. This also explains why Econeal can achieve sterilization even though it is bonded in the molecular chain of EA and cannot be released.

Hydrophilic analysis

Figure 8 reflects the correspondence between the static contact angle of EA-Econeal film to water and the

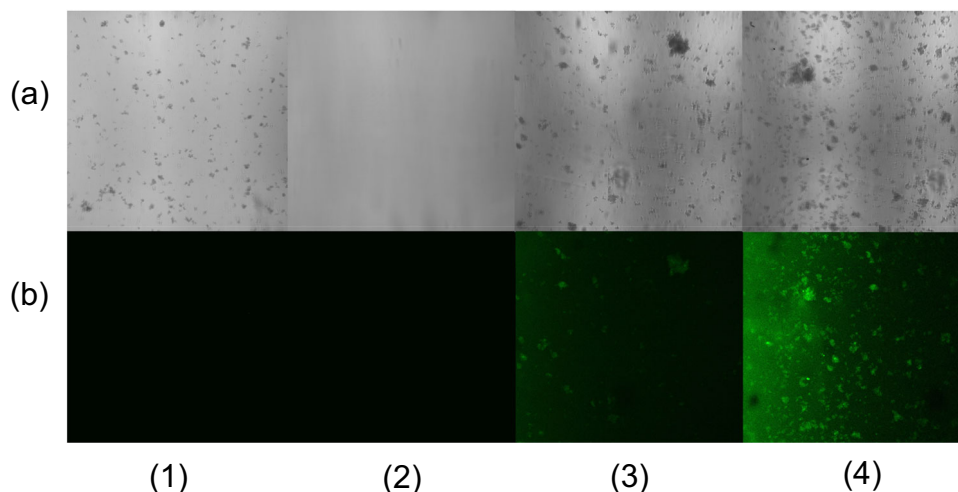


Fig. 7: CLSM images of *S. aureus* suspension and EA-Econea

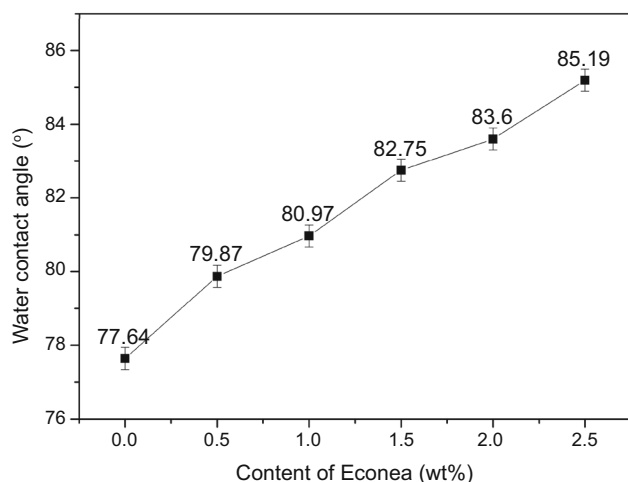


Fig. 8: Static contact angles to water of the EA-Econea film with different Econea content

Econea content. The figure shows that the contact angle continuously increases with the Econea content; the contact angle increases from 77.6° for pure EA to 85.2° for EA-Econea with 2.5 wt% Econea. Obviously, this is because of the hydrophobic F group in the Econea structure. As the hydrophobicity increases and the surface tension decreases, the microorganisms cannot easily adhere to the coating.

Antifouling performance

The antifouling ability of stainless steel plates, 50 × 50 cm² in size, coated with EA-Econea film was tested by immersing it in microorganism-rich river water. For comparison, row (a) and row (b) exhibit the immersion states of the stainless steel plate coated with pure EA and EA-Econea films, respectively, in Fig. 9. Judging from the immersion condition for one day, as shown in

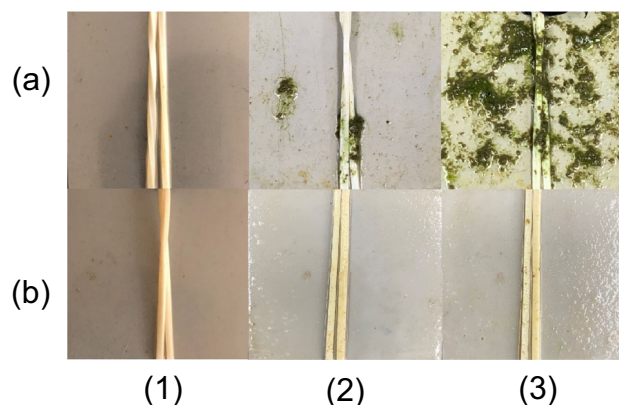


Fig. 9: Comparison of water immersion effect between stainless steel plates coated with pure EA and coated with EA-Econea, respectively

Figs. 9 (a, 1) and 9 (b, 1), dirt is present in an area on the surface of the stainless steel plate coated with pure EA film, whereas that coated with the EA-Econea film is very clean. As the immersion time is extended to 7 days and 60 days, as shown in Figs. 9 (a, 2) and (a, 3), respectively, the algae fouling on the stainless steel plate coated with pure EA film becomes increasingly serious. However, the stainless steel plates coated with the EA-Econea film, as shown in Figs. 9 (b, 2) and 9(b, 3), are very clean. Simultaneously, the adhesion of the coating film to the stainless steel plate is also very good, laying a good foundation for the use of the coating film in marine coatings.

Rows (a) and (b) display the images of stainless steel plates coated with pure EA and EA-Econea films, respectively; columns (1), (2), and (3) display images of the plate immersed in microorganism-rich river water for 1 day, 7 days, and 60 days, respectively

Conclusions

Herein, E-conea was used as the core functional group to perform the solution grafting reaction with the epoxy-containing acrylate copolymer EA to obtain the marine antifouling coating EA-E-conea. The grafting rate of E-conea combined with the EA molecular chain was 62.2%. Additionally, the EA-E-conea coating is homogeneous. The hydrophobicity of the coating increases with the bonding amount. Furthermore, when the mass fraction of E-conea involved in the reaction was 0.5%, the EA-E-conea coating had an antibacterial effect of >99.9% against *Escherichia coli* and *Staphylococcus aureus*, and the bacterial cell membranes were quickly destroyed by contact to achieve a sterilization effect. Therefore, sterilization can be achieved without breaking the E-conea and EA molecular chain bonding. This lays the foundation for the long-term performance of the coating. Moreover, the stainless steel plate coated with EA-E-conea was immersed in a microorganism-rich river for two months and it was not contaminated. This proves that killing bacteria on the surface of objects in water is beneficial for eradicating the basis of attachment of aquatic organisms. Hence, this study presents a possible solution for solving the seawater biological attachment problem that is plaguing the world.

Acknowledgments The authors acknowledge support from the National Natural Science Foundation of China (Grant No. 50573020).

References

1. Callow, ME, Fletcher, RL, "The Influence of Low Surface Energy Materials on Bioadhesion—A Review." *J. Int. Biodeteriorat. Biodegrad.*, **34** (3/4) 333–348 (1994)
2. Wahl, M, Marinee, I, "Fouling and Antifouling: Some Basic Aspects." *J. Mar. Ecol. Progress Ser.*, **58** (12) 175–189 (1989)
3. Lindholdt, A, Dam-Johansen, K, Oise, SM, et al. "Effects of Biofouling Development on Drag Forces of Hull Coatings for Ocean-Going Ships: A Review." *J. Coat. Technol. Res.*, **12** (3) 415–444 (2015)
4. Sweat, LH, Swain, GW, Hunsucker, KZ, et al. "Transported Biofilms and Their Influence on Subsequent Macrofouling Colonization." *J. Biofouling*, **33** (5) 433–449 (2017)
5. Schultz, MP, Bendick, JA, Holm, ER, et al. "Economic Impact of Biofouling on a Naval Surface Ship." *J. Biofouling*, **27** (1) 87–98 (2011)
6. Serrano, A, Sterner, O, Mieszkina, S, et al. "Nonfouling Response of Hydrophilic Uncharged Polymers." *J. Adv. Funct. Mater.*, **23** (46) 5706–5718 (2013)
7. Yebra, DM, Kiil, S, Dam-Johansen, K, "Antifouling Technology—Past, Present and Future Steps Towards Efficient and Environmentally Friendly Antifouling Coatings." *J. Process Organ. Coat.*, **50** (2) 75–104 (2004)
8. Cognetti, G, Maltagliati, F, Pretti, C, "Antifouling Coatings and Ecological Control in Marinas." *J. Mar. Pollut. Bull.*, **64** (2) 175–176 (2012)
9. Zhang, JW, Lin, CG, Wang, L, et al. "Study on the Correlation of Lab Assay and Field Test for Fouling-Release Coatings." *J. Progress Organ. Coat.*, **76** 1430–1434 (2013)
10. Li, PC, Cai, C, Long, YH, et al. "Cu₂O-Clay Composites with Sub-Micrometer-Sized Cu₂O Particles for Marine Antifouling Paints." *J. Coat. Technol. Res.*, **16** 25–30 (2019)
11. Galhenage, TP, Webster, DC, Moreira, AMS, et al. "Poly (ethylene) Glycol-Modified, a m Phiphilic, Siloxane-Polyurethane Coatings and Their Performance as Fouling-Release Surfaces." *J. Coat. Technol. Res.*, **14** (2) 307–322 (2017)
12. Rufin, MA, Ngo, BKD, Barry, M, et al. "Antifouling Silicones Based on Surface-Modifying Additive Amphiphiles." *J. Green Mater.*, **5** (1) 4–13 (2017)
13. Cheng, Z, Zhang, F, Sun, XY, et al. "Preparation and Properties of Amphiphilic Fluorinated Silicone Modified Acrylic Resin Coatings Containing PEG for Anti-fouling Application." *J. Acta Polymerica Sinica*, **8** 1112–1120 (2016)
14. Railkin, AI, *Marine Biofouling, Colonization Process and Defenses*. CRC Press (2004)
15. Kyei, SK, Darko, G, Akaranta, O, "Chemistry and Application of Emerging Ecofriendly Antifouling Paints: A Review." *J. Coat. Technol. Res.*, **17** 315–332 (2020)
16. Gatenholon, P, Holmstrom, C, Maki, JS, et al. "Toward Biological Antifouling Surface Coatings: Marine Bacteria Immobilized in Hydrogel Inhibit Barnacle Larvae." *J. Biofouling*, **9** 293–301 (1996)
17. Olga, F, Patricia, R, Joao, F, et al. "Biofouling Inhibition with Grafted E-conea Biocide: Toward a Nonreleasing Eco-Friendly Multiresistant Antifouling Coating." *J. ACS Sustain. Chem. Eng.*, **8** 12–17 (2020)
18. Yu, XY, Wang, K, Chen, ZT, et al. "Development and Performance of Copper Antifoulant-Free Marine Biocide Antifouling Paints." *J. Surface Technol.*, **46** (12) 11–17 (2017)
19. Thomas, KV, Brooks, S, "The Environmental Fate and Effects of Antifouling Paint Biocides." *J. Biofouling*, **26** (1) 73–88 (2010)
20. Nazli, M, Gamza, T, Levent, C, "RP-HPLC Optimization of E-conea by Using Artificial Neural Networks and Its Antifouling Performance on the Turkish Coastline." *J. Progress Organ. Coat.*, **77** (3) 627–635 (2014)
21. Loureiro, MV, Vale, M, De Schrijver, A, et al. "Hybrid Custom-Tailored Sol-Gel Derived Microscaffolds for Biocides Immobilization." *J. Microporous Mesoporous Mater.*, **261** 252–258 (2018)
22. Downs, RA, Dean, JR, Downer, A, et al. "Determination of the Biocide E-conea (R) in Artificial Seawater by Solid Phase Extraction and High Performance Liquid Chromatography Mass Spectrometry." *J. Sep.*, **4** 34 (2017)
23. Silva, ER, Ferreira, O, Ramalho, PA, et al. "Eco-Friendly Non-biocide-Release Coatings for Marine Biofouling Prevention." *J. Sci. Total Environ.*, **650** 2499–2511 (2019)

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